Shapes and angles — Quadrilaterals · Parallel lines · Angles in polygons

Give a reason for every angle you work out. Sketch each diagram described and mark the angles on it. ★ = challenge question.

A · QUADRILATERALS

1 Name all the quadrilaterals that have each property.

a All four sides equal in length.

b Diagonals that cross at right angles.

c Two pairs of parallel sides.

> **a** Square, rhombus. **b** Square, rhombus, kite. **c** Square, rectangle, rhombus, parallelogram.

2 A quadrilateral has two pairs of equal *adjacent* sides and exactly one pair of equal angles. What is it called?

> A kite.

3 One angle of a parallelogram is $50°$. Work out the other three angles, with reasons.

> Opposite angles of a parallelogram are equal: $50°$. Co-interior angles between the parallel sides add to $180°$: the other two are $130°$ each. (Check: $50 + 130 + 50 + 130 = 360$ ✓)

4 A kite has an angle of $110°$ at the top vertex and $70°$ at one of the side vertices. Work out the other two angles.

> The two side angles of a kite are equal: $70°$. Angles in a quadrilateral add to $360°$: bottom angle $= 360 - 110 - 70 - 70 = 110°$.

B · ANGLES AND PARALLEL LINES

5 Two parallel lines are crossed by a third line. One of the angles formed at the upper crossing is $85°$. Write down the size of **a** the alternate angle, **b** the corresponding angle, **c** the co-interior angle at the lower crossing, with a reason for each.

> **a** $85°$ — alternate angles are equal. **b** $85°$ — corresponding angles are equal. **c** $180 - 85 = 95°$ — co-interior angles add to $180°$.

6 A triangle sits between two parallel lines: its base lies on the lower line and its top vertex touches the upper line. The two angles at the top vertex, between the triangle's sides and the upper line, are $76°$ and $37°$. Work out the three angles of the triangle, with reasons.

> Base angles: $76°$ and $37°$ — each is the alternate angle of one of the angles at the top (parallel lines). Top angle: angles on a straight line, $180 - 76 - 37 = 67°$. Check: $76 + 37 + 67 = 180$ ✓ — the alternate angles trick is how we prove the triangle angle sum.

7 Two co-interior angles between a pair of parallel lines are $(2x + 20)°$ and $(3x + 10)°$. Find x and the two angles.

> Co-interior angles add to $180°$: $2x + 20 + 3x + 10 = 180 \Rightarrow 5x = 150 \Rightarrow x = 30$. Angles: $80°$ and $100°$.

C · ANGLES IN POLYGONS

8 **a** Write down the angle sum of a triangle, a quadrilateral, a pentagon and a hexagon.

b Five angles of a hexagon are $80°$, $90°$, $220°$, $150°$ and $120°$. Work out the sixth.

> **a** $180°$, $360°$, $540°$, $720°$ — each extra side adds $180°$ (one more triangle).
> **b** $720 - (80 + 90 + 220 + 150 + 120) = 720 - 660 = 60°$. (A $220°$ angle is reflex — the hexagon is not convex, but the rule still holds.)

9 The exterior angles of a quadrilateral are $85°$, $90°$, $40°$ and x.

a Work out x.

b Write down the sum of the interior angles.

> **a** Exterior angles of any polygon add to $360°$: $x = 360 - 215 = 145°$. **b** $360°$.

10 Work out **a** the exterior angle and **b** the interior angle of a regular 15-sided polygon.

> **a** $360 \div 15 = 24°$. **b** $180 - 24 = 156°$.

11 A regular polygon has exterior angles of $12°$. How many sides does it have?

> $360 \div 12 = 30$ sides.

12 Lines from the centre of a regular polygon to two neighbouring vertices make an angle of $8°$ at the centre.

a How many sides does the polygon have?

b Work out the two equal base angles of the isosceles triangle formed.

c Work out the interior angle of the polygon.

> **a** $360 \div 8 = 45$ sides. **b** $(180 - 8) \div 2 = 86°$ each. **c** Two base angles meet at each vertex: $2 \times 86 = 172°$ (check: $180 - \frac{360}{45} = 172$ ✓).

13 Which regular polygons tessellate (fit together with no gaps)? Explain using their interior angles.

> Shapes tessellate when their interior angle divides exactly into $360°$. Equilateral triangle $60°$ (6 meet at a point), square $90°$ (4), regular hexagon $120°$ (3) — these tessellate. Regular pentagon $108°$: $360 \div 108$ is not a whole number, so it does not; nor does any regular polygon with more than 6 sides.

Reasons: alternate / corresponding angles equal, co-interior add to 180° · polygon angle sum $(n-2) \times 180°$ · exterior angles always add to 360° · regular: exterior $= 360° \div n$.